Industrial Functional Programming [1]

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Contents

Database support

- Mnesia
- Tokio Cabinet, Kyoto Cabinet
- CauchDB
- MongoDB
- Hibari
- MySQL with ODBC
- RIAK
- etc.

(A)Mnesia

- Distributed DBMS
- Soft real-time
- Part of Erlang/OTP
- Easy to use: same data types
- Stores Erlang records

Mnesia

- Staring the shell: `erl -mnesia dir "location”`
- Starting: `mnesia:start()`
- Initialisation: `mnesia:create_schema/1`
- Creating tables: `mnesia:create_table/2`
- Selecting elements:
  `mnesia:select/3, match_object/*`
- Read/write elements: `mnesia:read/*, write/*`
- Transaction handling: `mnesia:transaction/1`

Web-servers

- Inets – Lightweight HTTP server built into Erlang
- Yaws (Yet Another Web Server) – HTTP server developed by Claes "Klacke" Wikstrom
- Mochiweb – HTTP server developed by Bob Ippolito/MochiMedia
- Webmachine – HTTP resource server developed by Basho Technologies (runs on Mochiweb under the hood
- Cowboy
- etc...
- and so many web frameworks

YAWS configuration, `yaws.conf`

```
<server localhost>
        port = 8000
        listen = 127.0.0.1
        docroot = /home/melinda/Desktop/Erlang_ora/
        dir_listings = true
#        auth_log = true
        statistics = true
#        <auth>
#                realm = foobar
#                dir = /
#                user = foo:bar
#                user = baz:bar
#        </auth>
</server>
```

index.yaws

```
<erl>

out(A) ->
    {ehtml, [erlcode:html_head("Welcome page!"),
             read(A),
             erlcode:html_end()]}.

read(A) ->
    Headers = (A#arg.headers)#headers.user_agent,
    [{p, [], "Browser data: " ++ io_lib:format("~p", [Headers])},
     {form, [{method, post}, {action, "/greetings.yaws"}],
            [{p, [], "Your name?"},
             {input, [{name, name}, {type, text},
                      {value, ""}, {size, 20}]},
             {input, [{type, submit}, {value, "OK"}]}]}].

</erl>
```

greetings.yaws

```
<erl>
out(A) ->
    L    = yaws_api:parse_post(A),
    Name = erlcode:key2value("name", L),
    Message =
        case Name of
            undefined -> "Welcome Anonymous!:)";
            _ -> "Greetings " ++ Name
        end,
    {ehtml, [erlcode:html_head(Message),
             [{h1, [], Message}], erlcode:html_end()]}.
</erl>
```

erlcode.erl

```
-module(erlcode).
-compile(export_all).

key2value(Key, PList) ->
    {value, {Key, Val}} = lists:keysearch(Key, 1, PList),
    Val.

html_head(PageTitle) ->
    ["<!DOCTYPE HTML PUBLIC \"-//W3C//DTD HTML 4.01 Transitional//EN\">
<html>
<head>
 <meta name=\"keywords\" content=\"Nortel Extranet VPN\">
 <title>" ++
     PageTitle ++
         "</title>
 <meta http-equiv=\"Content-Type\" content=\"text/html; charset=iso-8859-1\">
 <link rel=\"stylesheet\" type=\"text/css\" href=\"style.css\">
</head>

<body bgcolor=\"linen\">
<div align = center>"].

html_end() ->
    ["</div>
</body>
</html>"].
```